

The Organelle Matrix: Compare & Contrast

Distinguishing Sub-Cellular Structures Across Living Kingdoms

Name: _____ Date: _____ Class: _____ Score: _____ / 21

Task: Complete the matrix below by writing a tick (✓) if the sub-cellular structure is present, or a cross (✗) if it is absent in each cell type. (1 mark per box)

Sub-Cellular Structure	Animal (Eukaryote)	Plant (Eukaryote)	Bacteria (Prokaryote)
Nucleus Contains genetic material & chromosomes	☐	☐	☐
Cell Membrane Controls substance movement into/out of cell	☐	☐	☐
Mitochondria Site of aerobic respiration (releases ATP)	☐	☐	☐
Ribosomes Site of protein synthesis	☐	☐	☐
Chloroplasts Contains chlorophyll for photosynthesis	☐	☐	☐
Cell Wall Provides structural support & strength	☐	☐	☐
Plasmids Small circular rings of extra DNA	☐	☐	☐

Extension Challenge (+2): Name TWO structures found in ALL THREE cell types: 1. _____ 2. _____ Self-Check: ☐ Review ☐ Got It ☐ Can Teach

CUT ALONG LINE (2 IDENTICAL HANDOUTS PER A4 PAGE)

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The Organelle Matrix — Official Answer Key

GCSE Combined Science / GCSE Biology (Cell Biology Specification Standard)

Sub-Cellular Structure	Animal (Eukaryote)	Plant (Eukaryote)	Bacteria (Prokaryote)	Examiner Guidance & Biological Rationale
Nucleus	✓ Present	✓ Present	✗ Absent	Bacteria have no nucleus ; single circular DNA loop floats free in cytoplasm.
Cell Membrane	✓ Present	✓ Present	✓ Present	Found in all living cells to regulate molecular transport in/out.
Mitochondria	✓ Present	✓ Present	✗ Absent	Bacteria lack membrane-bound organelles; respiration occurs across the cell membrane.
Ribosomes	✓ Present	✓ Present	✓ Present	Found in all living cells for translation & protein synthesis (70S in prokaryotes, 80S in eukaryotes).
Chloroplasts	✗ Absent	✓ Present	✗ Absent	Found in photosynthetic green plant cells. (Note: underground plant cells like root hair cells lack them).
Cell Wall	✗ Absent	✓ Present	✓ Present	Plant cell walls are made of cellulose ; bacterial walls of peptidoglycan . Animal cells NEVER have a cell wall.
Plasmids	✗ Absent	✗ Absent	✓ Present	Small circular rings of accessory DNA characteristic of bacteria; often carry antibiotic resistance genes.

Extension Challenge Answer Key (+2 Marks): Structures present in ALL three cell types:
Any TWO of: **Cell Membrane** (1 mark), **Ribosomes** (1 mark), or **Cytoplasm** (1 mark).

⚠ **Critical GCSE Examiner Distinctions & Misconceptions:**

- 1. Bacterial DNA vs Nucleus:** A common student error is claiming bacteria have a nucleus. Emphasize that prokaryote means "before nucleus". Their chromosome is a loose circular strand plus plasmids.

3. Cell Wall Chemistry: Plant walls = cellulose (strengthens against turgor pressure). Bacterial walls = peptidoglycan. Animal cells never have cell walls.
- 2. "Do all plant cells have chloroplasts?":** Remind students that root hair cells and onion bulb cells do NOT have chloroplasts because they are underground and receive no sunlight.

4. Respiration in Bacteria: Even without mitochondria, bacteria still carry out respiration using electron transport enzymes on their plasma membrane.